

Project Paper

Spotify dataset

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Introduction

Music has always been an important part of human culture. Over time, the way people create, share, and listen to music has definitely changed based on some vital events. With digital platforms, we can now gather and study large amounts of music data.

Our project aims to explore how music has changed over the years using data visualization. We look at features like energy, danceability, acousticness, and popularity to see how they vary across time and genres.

We use three datasets. One shows average audio features by genre. Another tracks how these features change by year. The third gives detailed data on individual songs, including artist, release date, and popularity. These datasets help us understand both musical styles and trends over time.  Spotify dataset

Our goal is to present this information through simple and clear visualizations to show how music has evolved.

Methodology

To explore how music has evolved over time, we used a combination of visual and analytical tools. Our main approach was to rely on clear and accessible data visualizations that highlight trends and patterns in musical features.

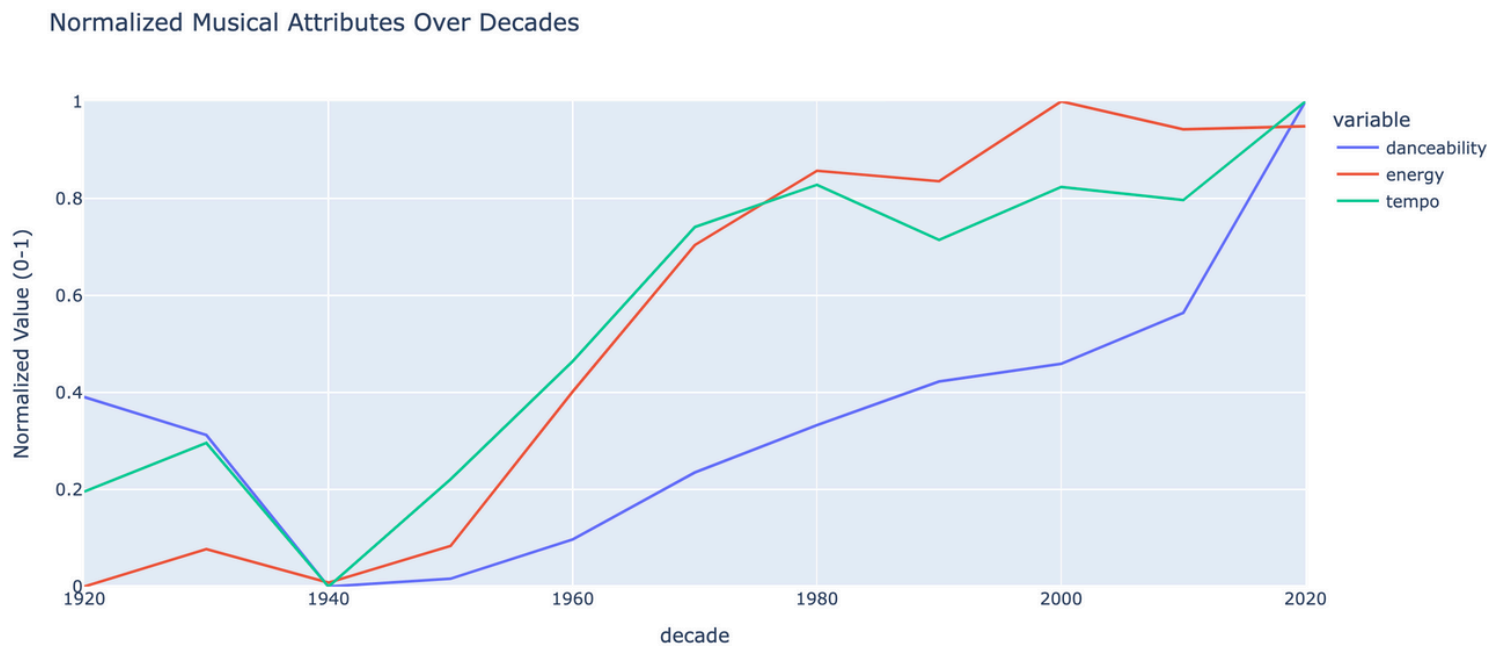
We used time series charts to observe how certain audio characteristics—such as energy, valence, and popularity—have changed across different decades. These charts helped us see general shifts in the mood and style of music over the years.

Distribution plots helped us understand the overall spread of values for key features, showing us how common or rare certain musical traits are. This was useful for identifying extremes and outliers within the data.

Beyond visualizations, we also applied some basic machine learning techniques to uncover deeper patterns. We used clustering methods, including K-Means, to group songs with similar characteristics. This helped us explore how different styles or genres naturally cluster together based on their audio features, without relying on labels.

Overall, our methodology combined data storytelling with analytical depth, allowing us to capture both the big picture and the hidden patterns in music data.

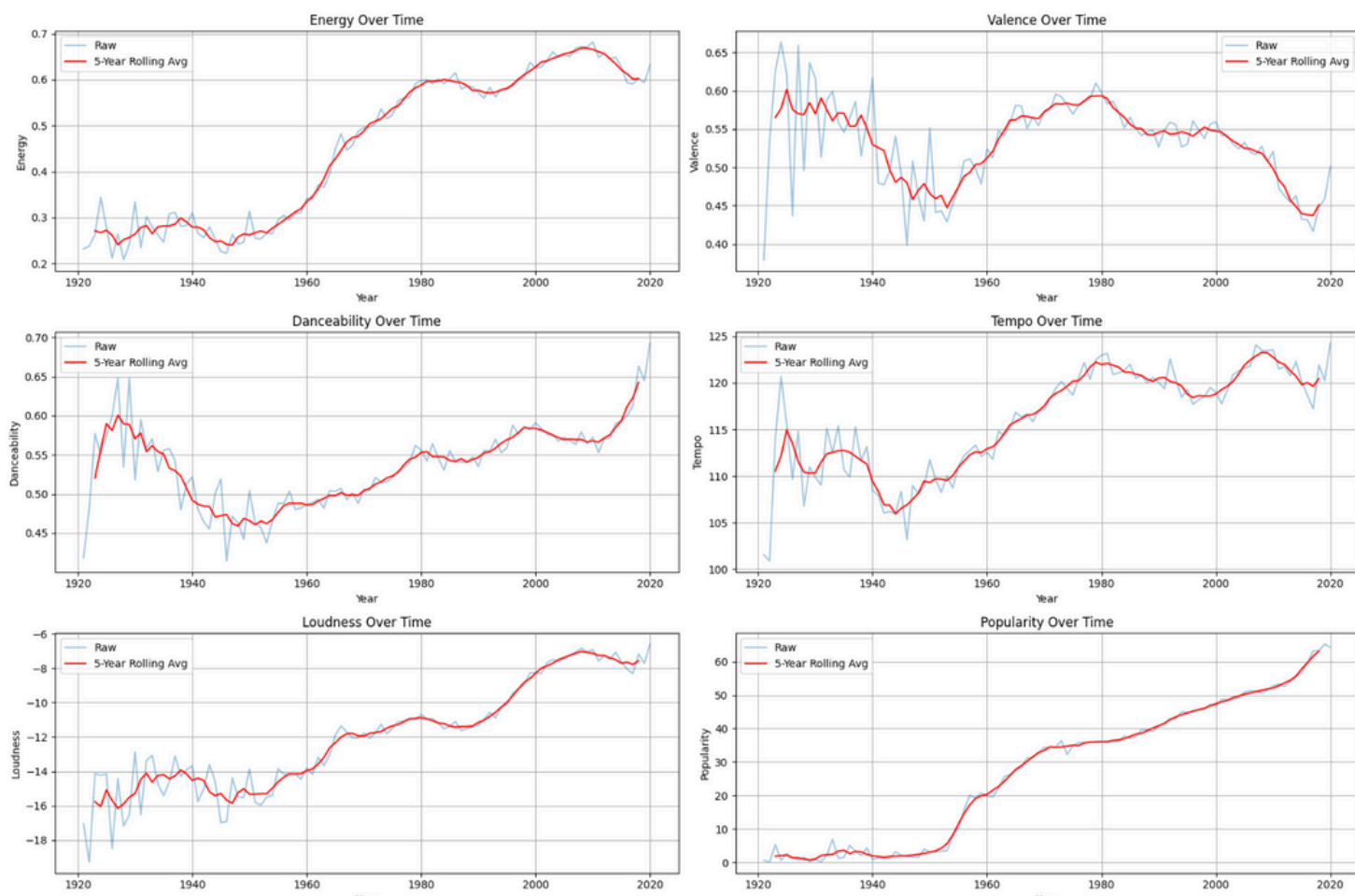
Visuals in the Paper



This visualizations illustrates the evolution of three core musical attributes—danceability, energy, and tempo—across different decades. Each attribute was normalized to a 0–1 scale, allowing for meaningful comparisons despite differences in their original scales.

This time series chart highlights how musical trends have shifted over the past century. We observe a notable increase in energy and tempo starting from the 1950s, reflecting the emergence of more upbeat and dynamic genres. Danceability, while initially lower, shows a steady upward trend, peaking in the most recent decades.

These patterns suggest broader changes in production styles, cultural influences, and listener preferences, emphasizing how music has adapted to evolving social contexts.



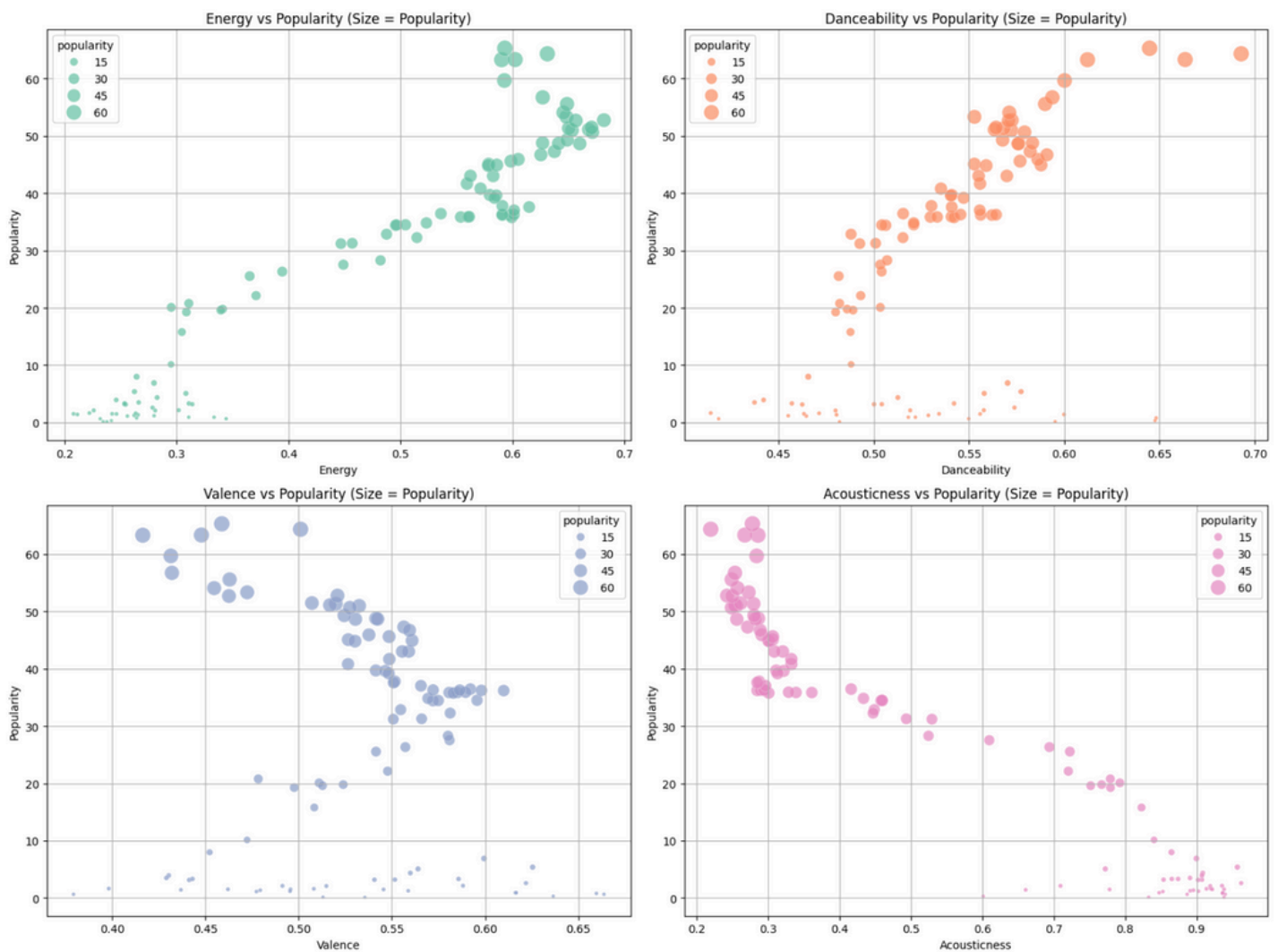
This set of line plots illustrates the evolution of six key musical attributes—energy, valence, danceability, tempo, loudness, and popularity—from the 1920s to the present. Each plot includes both the raw yearly values (light blue line) and a 5-year rolling average (solid red line), which serves to smooth short-term fluctuations and highlight long-term trends.

The rolling average is particularly effective in this context, as it reduces the visual noise caused by annual variability and allows for clearer interpretation of overarching changes in music characteristics. Line plots were chosen as the ideal format for this temporal data because they effectively show how values change over a continuous time axis and make trend comparisons straightforward.

This visualization reveals significant shifts in musical features across decades. For instance:

- Energy and loudness show a steady increase beginning around the 1950s, peaking in the early 2000s.
- Danceability dipped mid-century but has risen notably in recent decades.
- Valence, which measures musical positivity, shows a more complex trajectory, peaking in the 1960s and dipping more recently.
- Tempo exhibits moderate fluctuations, while popularity has clearly grown since the 1960s, likely reflecting improved data availability and broader audience engagement.

Overall, these plots provide a nuanced, data-driven view of how musical aesthetics and production qualities have evolved over the past century.

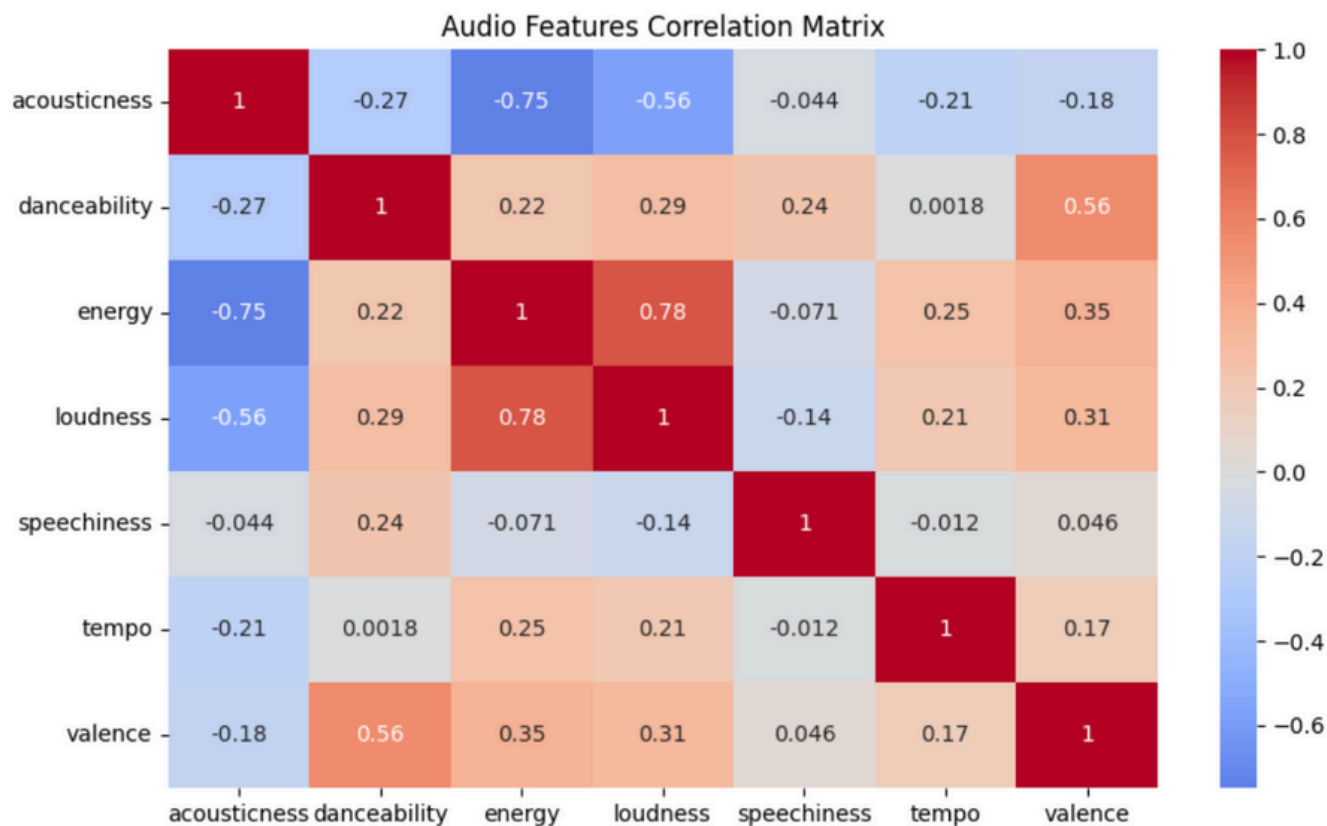


This set of scatter plots explores how four key musical attributes—energy, danceability, valence, and acousticness—relate to popularity. Each plot maps one feature against popularity, with the size of the points indicating the popularity score, enhancing the visual emphasis on more widely appreciated songs.

The scatter plot format is well-suited for this analysis, as it clearly illustrates the direction and strength of relationships between continuous variables. Moreover, using bubble size to reflect popularity adds a third dimension of insight, allowing for easy visual identification of high-impact songs.

- **Energy vs. Popularity:** There is a clear positive correlation, suggesting that more energetic songs tend to be more popular.
- **Danceability vs. Popularity:** A strong positive trend is visible here as well—songs with higher danceability are generally more popular, possibly due to their suitability for social and streaming contexts.
- **Valence vs. Popularity:** The relationship appears non-linear. While some higher-valence (happier) songs are popular, there is a noticeable spread, indicating that both emotionally positive and slightly melancholic songs can be well-received.
- **Acousticness vs. Popularity:** This relationship shows a strong negative correlation—songs with higher acousticness (i.e., more organic or unplugged) tend to be less popular, suggesting that mainstream popularity often favors more electronically produced music.

This visual approach effectively combines correlation analysis with intuitive bubble encoding, allowing for simultaneous evaluation of magnitude, trend, and density. It is especially useful for uncovering nuanced patterns that might be lost in simpler summaries or box plots.

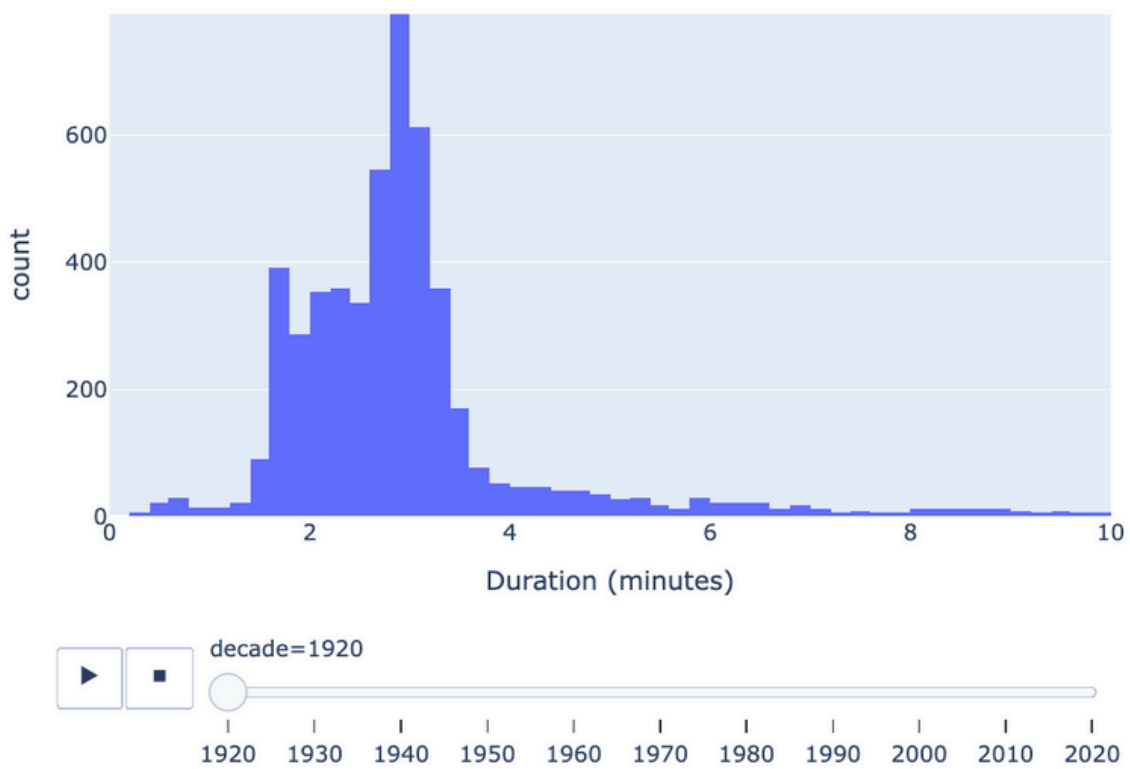


This heatmap visually represents the correlation between various audio features, revealing how different attributes interact. The correlation coefficients range from -1 to 1, with blue indicating negative correlations and red showing positive correlations.

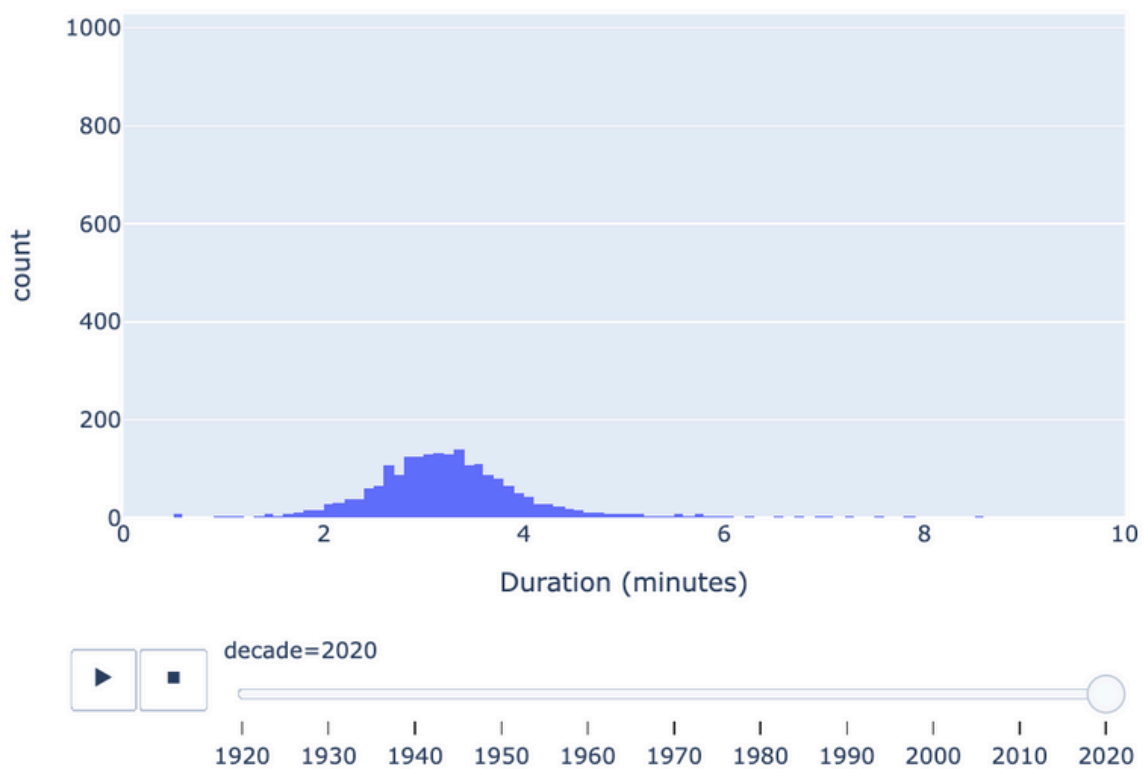
The heatmap is well-suited for this analysis because it allows for quick identification of relationships between continuous variables.

- Acousticness vs. Energy & Loudness: Strong negative correlations (-0.75 and -0.56). This suggests that highly acoustic songs tend to be quieter and less energetic.
- Danceability vs. Valence: A moderate positive correlation (0.56), indicating that more danceable songs are generally happier.
- Energy vs. Loudness: Strong positive correlation (0.78), meaning that louder songs tend to have higher energy.
- Speechiness: Displays weak correlations with other features, suggesting that speech-heavy tracks do not necessarily align with energy, danceability, or valence.
- Tempo: Shows weak correlations across the board, implying that the speed of a song does not strongly dictate other musical characteristics.

Track Duration Evolution by Decade



Track Duration Evolution by Decade



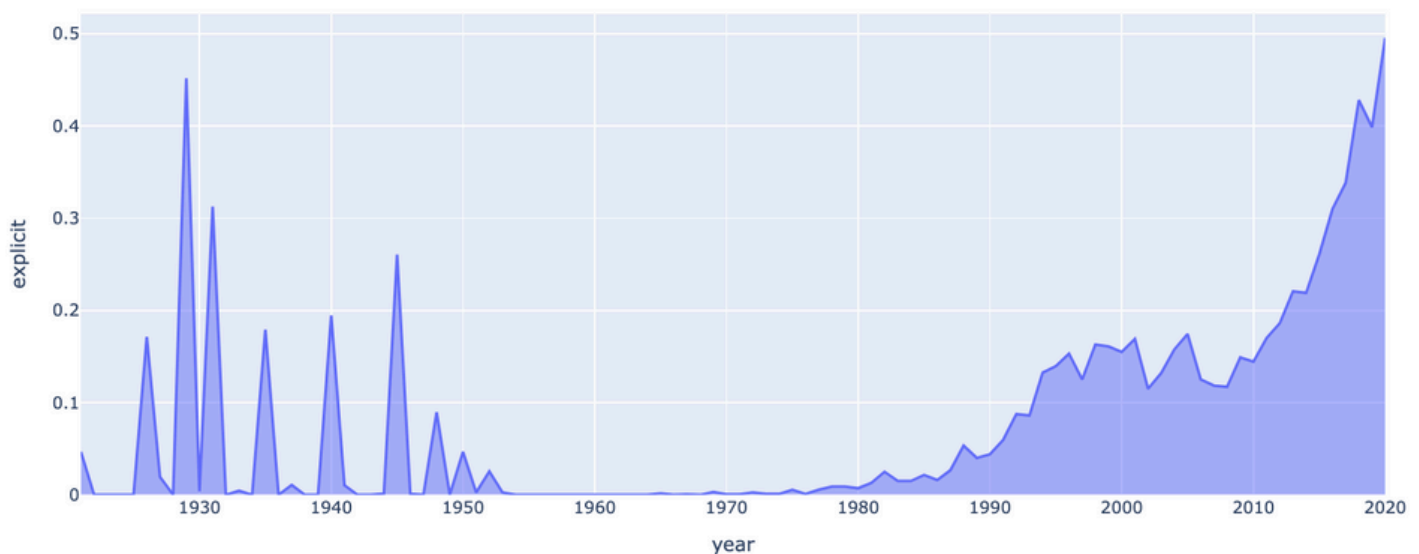
This animated histogram visualizes changes in the distribution of song durations from the 1920s to the 2020s, providing insight into shifts in music production trends. The x-axis represents track length (0–10 minutes), while the y-axis indicates song frequency.

The histogram format effectively captures temporal changes in song duration, showing how preferences and industry trends evolved. By animating across decades, it allows for an intuitive comparison between different eras, emphasizing long-term patterns rather than individual observations.

Key Observations

- 1920s: Songs exhibit a broader duration range, with many tracks surpassing 5 minutes. This reflects early recording constraints and musical styles that favored extended compositions.
- 2020s: The peak duration lies between 2–4 minutes, indicating a shift toward shorter, more radio-friendly songs, likely driven by modern consumption patterns such as streaming and social media.
- General Trend: The average song length has decreased over time, reinforcing the idea that contemporary music production prioritizes accessibility and engagement.

Share of Explicit Tracks Over Time



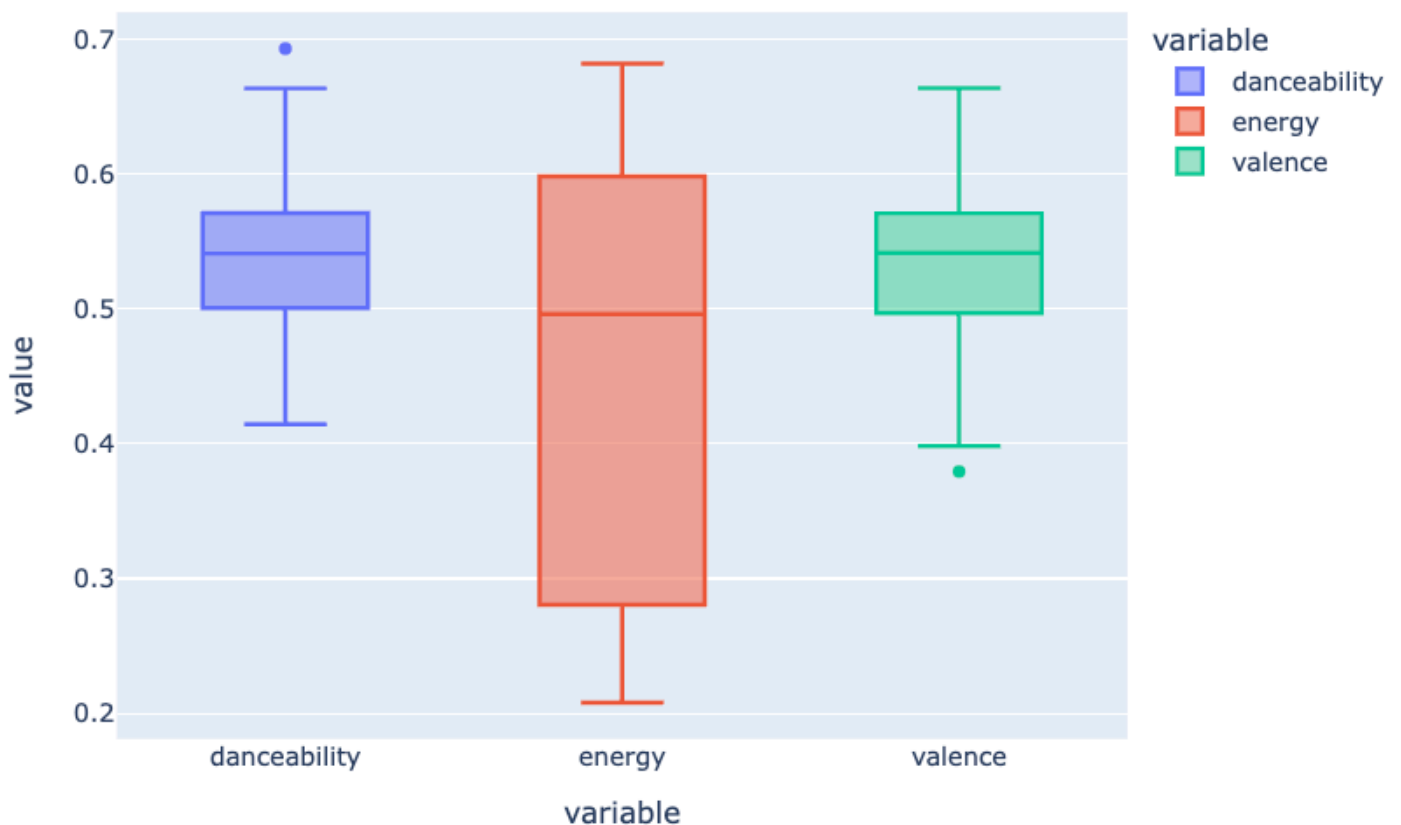
This area plot illustrates the yearly trend in the proportion of tracks labeled as explicit from 1920 to 2020.

The area plot is well-suited for this analysis as it emphasizes the overall progression and magnitude of explicit content prevalence. The continuous representation allows for clear identification of shifts and trends, making it easier to compare different decades.

- 1920s–1950s: The proportion of explicit tracks remains low and relatively stable, indicating minimal presence of explicit content in commercially released songs.

- 1960s–1980s: A slight increase occurs, coinciding with cultural shifts and evolving lyrical themes in music.
- 1990s–2000s: There is a noticeable rise, reflecting changing industry norms and the mainstream acceptance of explicit content.
- 2010s–2020s: The highest proportion of explicit tracks is observed, suggesting a peak in explicit music production, likely driven by streaming services and evolving listener preferences.

Feature Stability Over Time



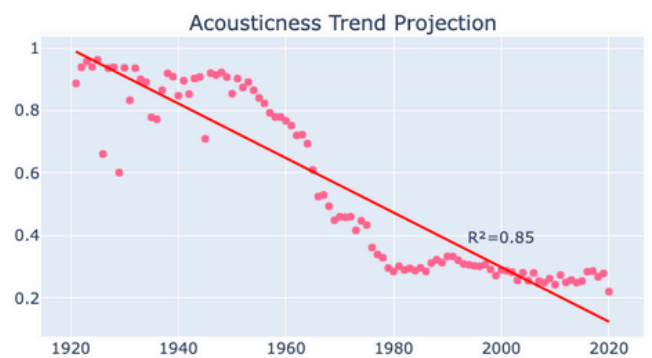
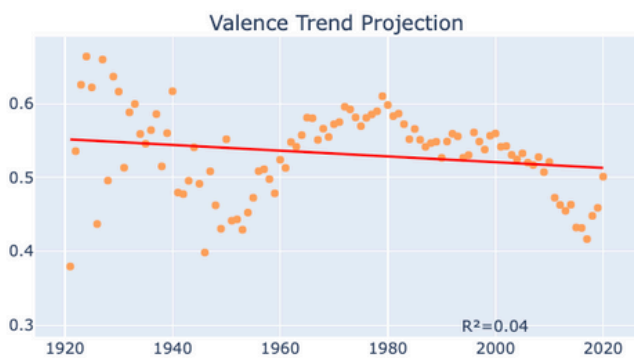
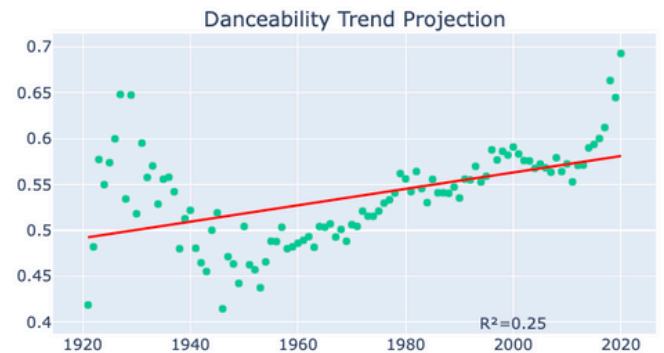
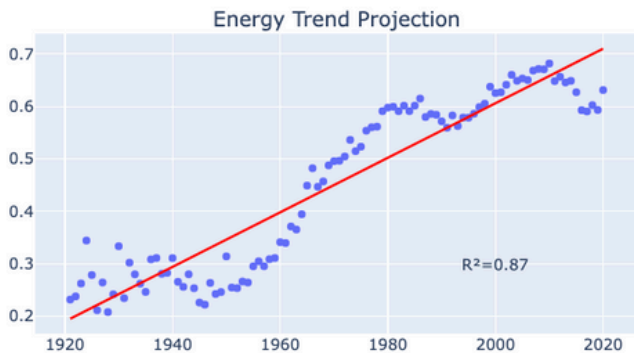
Box plots are particularly useful for tracking changes over time because they summarize key aspects of the distribution, including the median, quartiles, and outliers. This format helps reveal overall trends while also identifying variations that might be lost in simpler summaries.

Key Observations

- Danceability (blue): This feature appears relatively stable, with a consistent median value over time. However, outliers suggest occasional shifts in musical style, likely influenced by genre-specific trends.
- Energy (red): Displays a broader spread, with noticeable fluctuations. The interquartile range suggests variability in production styles, with some periods favoring high-energy compositions while others lean toward softer sounds.

- Valence (green): Shows a more dispersed pattern, indicating variation in emotional tone across different decades. Peaks and dips suggest shifts in musical sentiment, possibly reflecting cultural or industry trends.

Feature Trend Projections with OLS Trendlines



This set of scatter plots visualizes the trend projections of four key musical attributes—Energy, Danceability, Valence, and Acousticness over time.

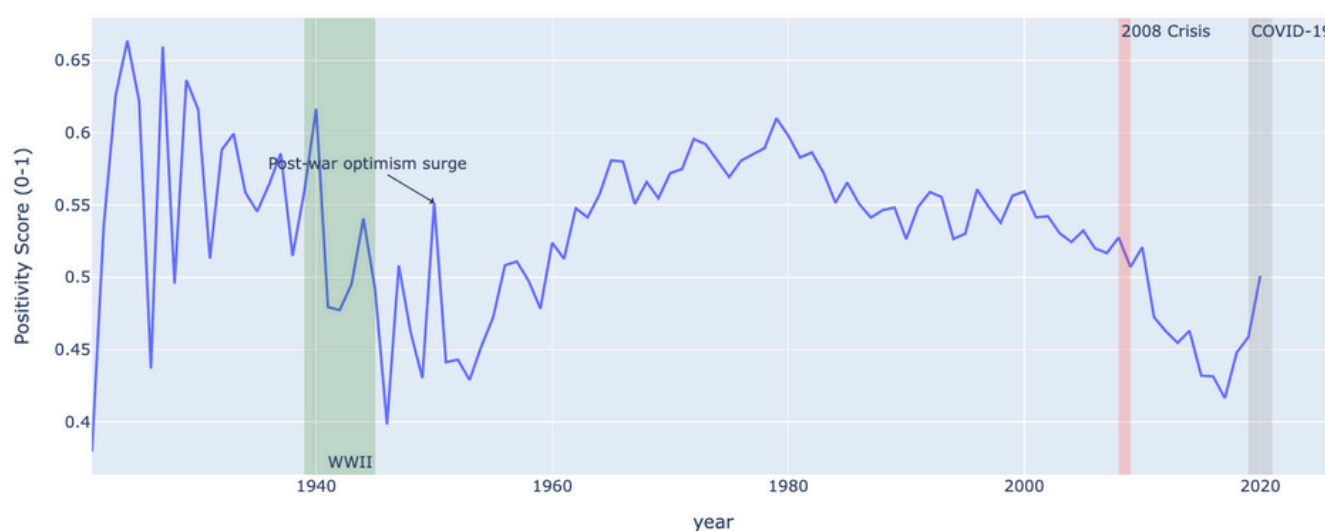
Scatter plots with OLS trendlines provide a clear representation of how continuous variables evolve over time. The inclusion of R-squared values helps assess the strength of each trend, ensuring that the visualizations offer meaningful insights rather than arbitrary fluctuations.

Key Observations

- Energy Trend Projection:
 - Displays a strong positive correlation ($R^2 = 0.87$), suggesting that songs have consistently become more energetic over the decades.
 - This could be attributed to advancements in production techniques and the demand for high-energy tracks in popular music.
- Danceability Trend Projection:
 - Shows a moderate positive correlation ($R^2 = 0.25$), indicating a slight increase in danceability.
 - While not as pronounced as energy, the shift may reflect evolving music trends favoring rhythm-driven compositions.

- **Valence Trend Projection:**
 - Exhibits a weak negative correlation ($R^2 = 0.04$), implying that while there has been a slight decline in overall musical positivity, the trend is not strong enough to suggest a definitive shift.
 - The wide variance suggests that both upbeat and melancholic songs remain popular across different eras.
- **Acousticness Trend Projection:**
 - Shows a strong negative correlation ($R^2 = 0.85$), revealing a significant decrease in acoustic elements in mainstream music.
 - This aligns with the rise of electronic production methods, signaling a shift away from organic instrumentation.

Musical Positivity Through Historical Events

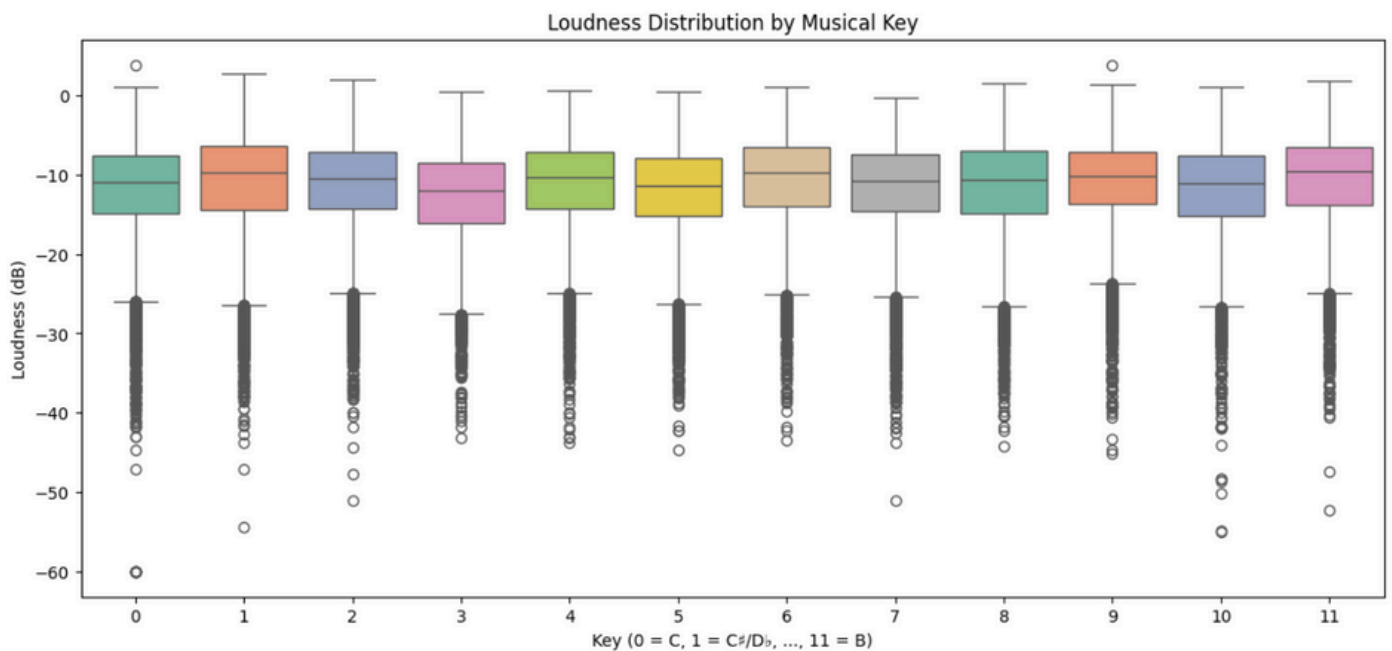


This line chart visualizes the progression of valence (musical positivity) from 1920 to 2020, integrating annotations that mark significant historical events such as World War II (1939–1945), the 2008 financial crisis, and the COVID-19 pandemic. By doing so, the visualization highlights how global upheavals may have influenced the emotional tone of popular music.

The line chart is particularly well-suited for tracking trends over time, as it allows for the identification of gradual shifts in musical sentiment. The colored vertical rectangles emphasize key historical periods, visually associating these events with fluctuations in positivity.

- **Pre-WWII (1920s–1930s):** The valence levels remain relatively high, indicating that popular music maintained a generally upbeat tone before the onset of war.
- **WWII Period (1939–1945):** A decline in valence suggests that music became less positive, potentially reflecting the somber cultural climate of the time.
- **Post-War Optimism (1950s):** A significant surge in musical positivity coincides with economic recovery and a cultural shift toward optimism, reflecting a renewed enthusiasm in popular music.

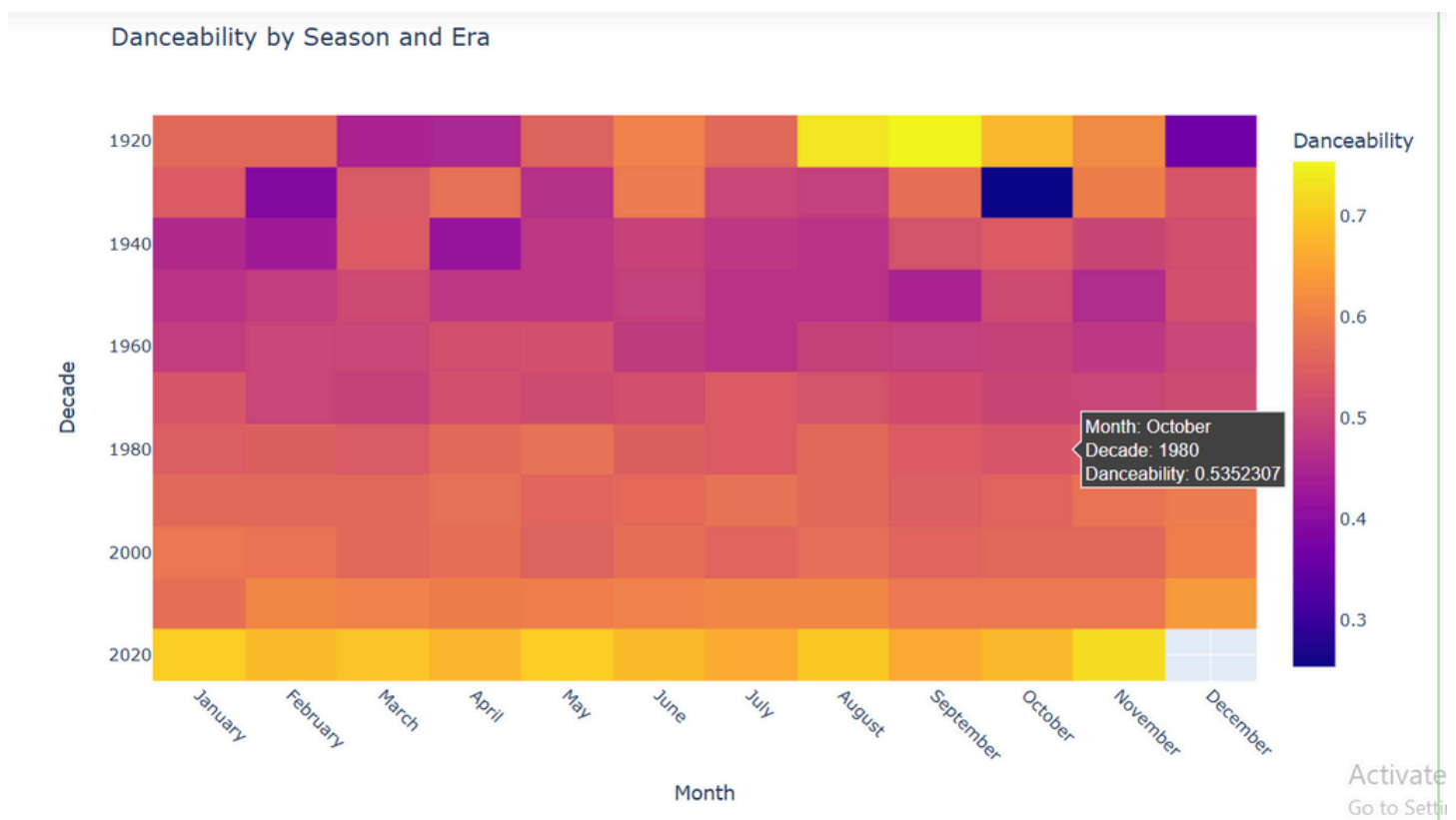
- 2008 Financial Crisis: A moderate dip in valence occurs around this period, possibly mirroring the economic uncertainty and societal stress.
- COVID-19 Pandemic (2020): A sharp decline in musical positivity is observed, reinforcing the idea that global crises impact lyrical themes and compositional choices.



This box plot visualizes the distribution of loudness (in decibels) across the 12 musical keys, labeled from 0 (C) to 11 (B).

Box plots are well-suited for analyzing distributions and identifying patterns across categorical variables. In this case, the visualization highlights how different keys may have varying loudness characteristics, helping to determine whether certain musical keys are generally louder or softer.

- Key C (0) and Key G (7) exhibit higher median loudness values, suggesting that songs in these keys tend to be louder on average.
- Key F# (6) and Key Bb (10) show a lower median loudness, indicating softer compositions in these tonalities.
- Outliers are present in multiple keys, reflecting instances of extreme loudness variations that may result from genre differences, production techniques, or mastering choices.
- Consistency vs. Variability: Some keys demonstrate a tighter spread (e.g., C and D), indicating more uniform loudness levels. Others show wide variations (e.g., E and A), suggesting greater diversity in production styles.

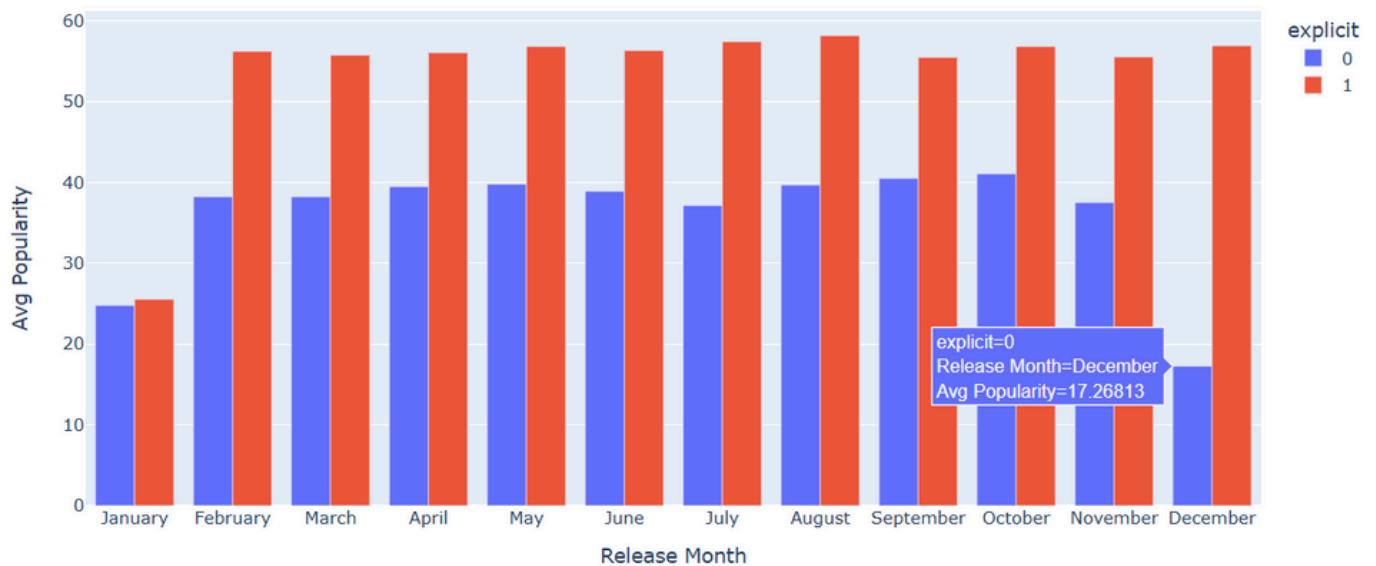


This heatmap visualizes the average danceability of tracks across different decades and months of release, revealing both long-term shifts in musical trends and potential seasonal influences.

Heatmaps are particularly useful for displaying dense temporal data, as they allow for clear comparisons across different time periods and seasonal variations. The month-wise breakdown within each decade ensures that both overarching trends and potential cyclical patterns are visible.

- **Overall Increase in Danceability:** The heatmap suggests a general upward trend in danceability over the decades, reflecting evolving production techniques and audience preferences favoring rhythm-driven music.
- **1980s & 2000s Peak:** Higher danceability scores are visible in these decades, likely influenced by the rise of disco, electronic music, and later streaming-era pop production favoring high-energy beats.
- **Seasonal Variation:**
 - Danceability tends to increase in summer months (June–August) across multiple decades, possibly driven by seasonal music trends and festival-driven releases.
 - December and January show relatively lower danceability scores, indicating a preference for more mellow tracks during holiday and winter seasons.
- **1920s–1950s Lower Danceability:** Earlier decades display more muted danceability values, aligning with historical musical styles that prioritized instrumentation over rhythmic emphasis.

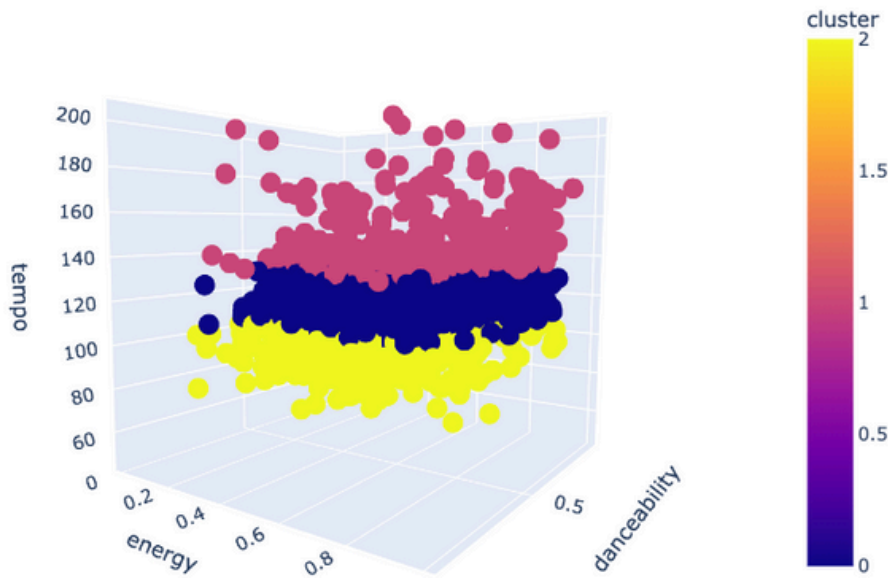
Optimal Release Timing



This bar chart illustrates the average popularity of music tracks, categorized by release month and whether they are explicit or non-explicit.

Bar charts are well-suited for comparative analysis, enabling a clear distinction between explicit and non-explicit music trends across different months.

- Explicit tracks consistently show higher average popularity across all months, suggesting a broader mainstream appeal or higher engagement.
- Peak popularity for explicit tracks is observed in February, March, and July, potentially aligning with industry release cycles or seasonal listening behaviors.
- Non-explicit tracks display lower average popularity, with the lowest values observed in December, possibly reflecting a preference for holiday-themed or mellower music toward year-end.
- General Trend: The disparity in popularity suggests that explicit music performs better on average, regardless of season, reinforcing its dominance in modern streaming trends.

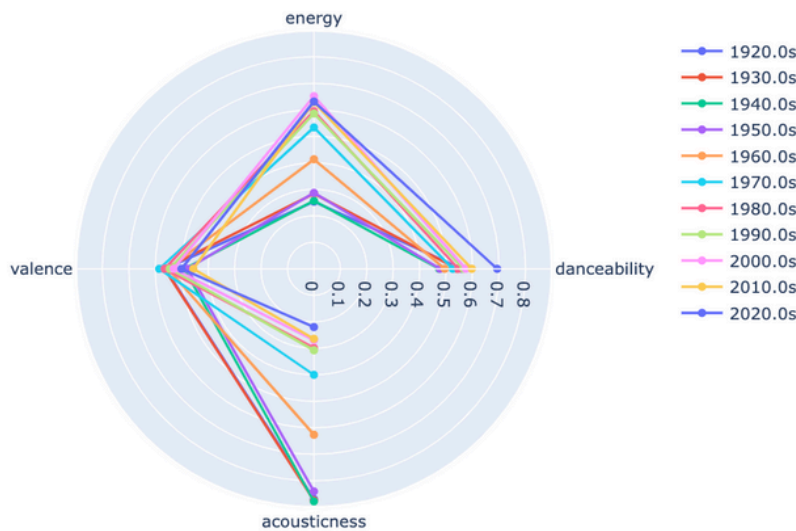


This 3D scatter plot visualizes how music tracks are grouped using K-Means clustering, based on three musical attributes: danceability, energy, and tempo. Each point represents a track, with its position determined by its values for these features. The color coding indicates different clusters, helping to uncover patterns in music genres.

K-Means clustering is ideal for segmenting data into meaningful groupings, allowing us to observe natural separations in musical styles. The 3D format enhances the interpretability of these relationships, enabling the viewer to explore variations across multiple dimensions simultaneously.

- Distinct clusters emerge, indicating that genres tend to share similar danceability, energy, and tempo characteristics.
- High-energy and high-tempo tracks form their own group, likely representing genres such as EDM or rock.
- Lower-energy, mid-tempo tracks cluster separately, potentially correlating with genres such as jazz, acoustic, or classical.
- Overlap between clusters suggests that some genres exhibit a blend of characteristics, highlighting musical diversity.

Attribute Evolution by Decade (Radar Chart)



This radar chart illustrates how key musical attributes danceability, energy, valence, and acoustiveness—have evolved from 1920 to 2020. Each line represents the average values of these attributes for a specific decade, providing a clear comparison of trends across time.

Radar charts are well-suited for multi-variable comparisons across different time periods. By plotting multiple attributes on the same scale, this visualization highlights how musical styles shift collectively rather than in isolation.

- **Energy & Danceability Increase:** Over time, music has become more energetic and danceable, reflecting changes in production techniques and audience preferences for high-intensity tracks.
- **Valence Variability:** While some decades exhibit higher musical positivity, others show dips, particularly in periods of historical or cultural shifts.
- **Decline in Acoustiveness:** There is a clear downward trend, suggesting that modern music increasingly favors electronic production over organic, unplugged elements.

Findings & Insights

Through our analysis of various musical attributes, we observe clear trends in how music has evolved over the decades. These shifts appear to be influenced by broader cultural movements, technological advancements, and psychological factors that shape listener preferences. Below, we outline key assumptions based on the observed trends and their potential real-world implications.

1. Increasing Energy and Danceability Reflecting Cultural & Psychological Shifts

One of the most noticeable trends is the rise in energy and danceability over time. The shift toward high-energy music suggests an increasing demand for more engaging and stimulating listening experiences. Possible reasons for this trend include:

- The influence of digital production—advancements in sound engineering and synthesizers made energetic compositions more accessible.
- The psychological need for stimulation—high-energy music tends to enhance motivation and physical engagement, making it well-suited for modern lifestyles.
- Changing consumption patterns—music is now consumed in short bursts through playlists, social media, and workouts, reinforcing the preference for highly rhythmic and energetic tracks.

This transformation implies that modern music prioritizes intensity, aligning with how audiences engage with entertainment media. The dominance of electronic and dance-heavy genres supports the idea that music functions increasingly as a tool for emotional activation and engagement, rather than passive listening.

2. The Decline of Acousticness Signaling a Shift Away from Traditional Music Styles

Over the decades, we observe a steady decline in acoustic elements, suggesting that organic instrumentation has become less prominent in mainstream music. Several factors may contribute to this:

- The rise of electronic music production—artists now rely heavily on digital tools rather than traditional instruments.
- Industry efficiency—electronic production allows for faster, more cost-effective music creation, reducing reliance on live instrumentation.
- Listener adaptation—modern audiences associate polished, digitally engineered sounds with mainstream appeal.

This shift suggests that music has transitioned into a largely electronic domain, reinforcing industry preferences for streamlined production methods. It raises questions about the future role of acoustic compositions and whether they will remain relevant in mainstream consumption.

3. Growth in Explicit Content as a Reflection of Cultural and Social Changes

A significant rise in explicit lyrics aligns with broader changes in societal values. Music appears to have become a space for unfiltered storytelling, where artists feel increasingly free to express raw, intense emotions. Possible explanations for this trend include:

- The shift toward authenticity in music—modern audiences value openness and artistic honesty, making explicit content more acceptable.
- The influence of hip-hop and countercultural movements—genres emphasizing direct, confrontational storytelling have normalized explicit expression.

- Digital platforms removing previous content restrictions—streaming services allow artists to release unrestricted lyrics, bypassing radio censorship.

This trend implies that explicit content is no longer a commercial barrier—in fact, it may contribute to audience engagement. Music functions as a reflection of social realities, reinforcing the idea that contemporary audiences seek music that mirrors their lived experiences.

4. Shorter Songs Driven by the Evolution of Listening Habits

Music tracks have become significantly shorter, shifting from extended compositions to concise formats. Several factors may explain this trend:

- The rise of streaming platforms—shorter songs align with digital consumption patterns, encouraging repeat listens.
- The psychological preference for instant gratification—modern listeners engage better with fast-paced, hook-driven music.
- Industry monetization models—streaming revenue favors tracks designed for high replayability, incentivizing condensed compositions.

This shift suggests that music is adapting to digital consumption behavior, where audiences engage in fragmented, rapid interactions with media. The trend toward shorter, highly engaging tracks reflects changing listener habits that prioritize immediacy and accessibility.

5. Seasonal & Historical Influences on Music Composition

Music consumption appears to follow seasonal and historical patterns, reinforcing the idea that musical attributes align with societal moods.

- Summer months show higher danceability, suggesting a preference for upbeat, high-energy tracks during warm seasons.
- Winter months favor mellow compositions, indicating a shift toward more introspective listening in colder periods.
- Major global events shape musical positivity (valence)—crises such as WWII, financial downturns, and the COVID-19 pandemic correlate with drops in musical optimism.

These patterns imply that music reflects and responds to societal shifts, functioning both as an emotional outlet and a cultural indicator. The ability of music to adapt to collective moods suggests that artists and industry professionals must remain aware of external influences when crafting compositions.

The trends we have observed suggest that music is undergoing constant transformation, shaped by listener behaviors, psychological needs, and industry forces. The transition toward high-energy, electronically produced, shorter, and explicit content-driven tracks reflects broader cultural shifts that prioritize engagement, emotional resonance, and accessibility. As music continues to evolve, understanding these dynamics will be crucial for artists, producers, and industry stakeholders navigating the ever-changing landscape of musical

consumption.

Conclusion & Recommendations

The evolution of music over the decades reflects broader shifts in production techniques, listener engagement, and societal changes. Our analysis highlights how technological advancements, psychological influences, and commercial strategies have driven trends such as higher energy levels, increased explicit content, the decline of acoustic instrumentation, shorter song formats, and seasonal listening patterns. These findings emphasize the interconnected nature of cultural adaptation and industry priorities.

To remain competitive, artists, producers, and industry strategists must anticipate these transformations, ensuring that music continues to align with evolving technologies, audience preferences, and global shifts.